

2 A computer game is being developed using object-oriented programming.

One element of the game is a balloon. This is designed as the class `Balloon`.

The class has the following attributes and methods.

Balloon	
<code>Health : INTEGER</code>	The health of the balloon
<code>Colour : STRING</code>	The colour of the balloon
<code>DefenceItem : STRING</code>	The item the balloon uses to defend itself
<code>Constructor()</code>	Initialises the defence item and colour using the parameters Initialises health to 100
<code>ChangeHealth()</code>	Takes the change as a parameter and adds this to the health
<code>GetDefenceItem()</code>	Returns the defence item of the object
<code>CheckHealth()</code>	If the health is 0 or less, it returns <code>TRUE</code> , otherwise it returns <code>FALSE</code>

- (a) The constructor takes the name of the defence item and the balloon's colour as parameters and sets these to the attributes. The health is initialised to 100.

Write program code to declare the class `Balloon` and its constructor. Do not write any other methods.

Use your language appropriate constructor.

All attributes should be private. If you are writing in Python include attribute declarations using comments.

Save your program as **Question2_J2022**.

Copy and paste the program code into **part 2(a)** in the evidence document.

[5]

- (b) The get method `GetDefenceItem()` returns the defence item of the object.

Amend your program code to include the get method `GetDefenceItem()`.

Save your program.

Copy and paste the program code into **part 2(b)** in the evidence document.

[2]

- (c) The object's method `ChangeHealth()` takes an integer number as a parameter and adds this to the health attribute of the object.

Amend your program code to include the method `ChangeHealth()`.

Save your program.

Copy and paste the program code into **part 2(c)** in the evidence document.

[2]

- (d) The object's method `CheckHealth()` returns `TRUE` if the health of the object is 0 or less (no health remaining) and returns `FALSE` otherwise (health remaining).

Amend your program code to include the method `CheckHealth()`.

Save your program.

Copy and paste the program code into **part 2(d)** in the evidence document.

[2]

- (e) Amend the main program to:

- take as input a defence item and colour from the user
- create a new balloon with the identifier `Balloon1` using the data input.

Save your program.

Copy and paste the program code into **part 2(e)** in the evidence document.

[3]

- (f) The function `Defend()`:

- takes a balloon object as a parameter
- takes as input the strength of an opponent from the user
- uses the `ChangeHealth()` method to subtract the strength input from the object's health
- outputs the defence item of the balloon
- checks the health of the object and outputs an appropriate message if it has no health remaining, or if it has health remaining
- returns the amended balloon object.

Write program code to declare the function `Defend()`.

Save your program.

Copy and paste the program code into **part 2(f)** in the evidence document.

[8]

(g) (i) Amend the main program to call the function `Defend()`.

Save your program.

Copy and paste the program code into **part 2(g)(i)** in the evidence document.

[2]

(ii) Test your program using the following inputs:

- balloon defence method "Shield"
- balloon colour "Red"
- strength of opponent 50

Take a screenshot to show the output.

Copy and paste the screenshot into **part 2(g)(ii)** in the evidence document.

[1]